ClipSAT : Math for AI & Data Science

Unit 3.01: Gradient Descent & Optimization

Name: _____

Q1. In gradient descent, a weight currently has value w = 10 . If the gradient of the loss is frac{partial L}{partial w} = 4 and the learning rate is eta = 0.1 , find the updated weight w' .

[ ] A: 10.40
[ ] B: 9.60
[ ] C: 6.00
[ ] D: 9.90

Q2. In gradient descent, a weight currently has value w = 20 . If the gradient of the loss is frac{partial L}{partial w} = 8 and the learning rate is eta = 0.2 , find the updated weight w' .

[ ] A: 12.00
[ ] B: 21.60
[ ] C: 18.40
[ ] D: 19.80

Q3. In gradient descent, a weight currently has value w = 30 . If the gradient of the loss is frac{partial L}{partial w} = 12 and the learning rate is eta = 0.3 , find the updated weight w' .

[ ] A: 18.00
[ ] B: 33.60
[ ] C: 26.40
[ ] D: 29.70

Q4. In gradient descent, a weight currently has value w = 40 . If the gradient of the loss is frac{partial L}{partial w} = 16 and the learning rate is eta = 0.4 , find the updated weight w' .

[ ] A: 24.00
[ ] B: 46.40
[ ] C: 33.60
[ ] D: 39.60

Q5. In gradient descent, a weight currently has value w = 50 . If the gradient of the loss is frac{partial L}{partial w} = 20 and the learning rate is eta = 0.5 , find the updated weight w' .

[ ] A: 30.00
[ ] B: 60.00
[ ] C: 40.00
[ ] D: 49.50

Q6. In gradient descent, a weight currently has value w = 60 . If the gradient of the loss is frac{partial L}{partial w} = 24 and the learning rate is eta = 0.6 , find the updated weight w' .

[ ] A: 36.00
[ ] B: 74.40
[ ] C: 45.60
[ ] D: 59.40

Name: _____

Q7. Consider the scalar loss function Lw = 2w^2 - 8w + 5 . Find the analytical gradient nabla Lw evaluated at w = 2 .

[ ] A: nabla L2 = 16

[ ] B: nabla L2 = 8

[ ] C: nabla L2 = 0

[ ] D: nabla L2 = -4

Q8. Consider the scalar loss function Lw = 3w^2 - 8w + 5 . Find the analytical gradient nabla Lw evaluated at w = 4 .

[ ] A: nabla L4 = 16

[ ] B: nabla L4 = 24

[ ] C: nabla L4 = 32

[ ] D: nabla L4 = 4